

Math 1512 Calculus 1 Review

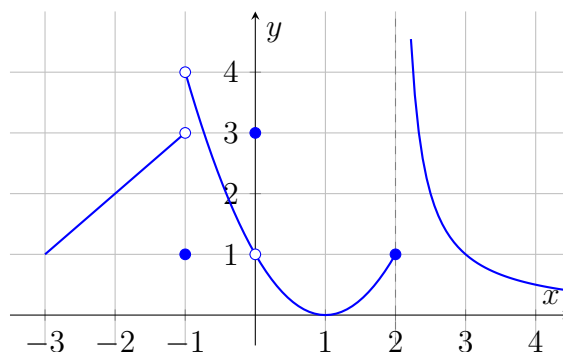
Fall 2026

1. Formulas to know.

- (a) State the definition of f is continuous at $x = a$.
- (b) Write $f'(a)$, the derivative of f at $x = a$, as a limit in two ways.
- (c) Give both parts of the Fundamental Theorem of Calculus.

2. Limits from a graph.

- (a) Let g be the function graphed below (a filled dot is a point on the graph, an open dot is not). Find $g(-1)$, $\lim_{x \rightarrow -1^-} g(x)$, $\lim_{x \rightarrow -1^+} g(x)$, $\lim_{x \rightarrow -1} g(x)$, then $g(0)$ and $\lim_{x \rightarrow 0} g(x)$, then $\lim_{x \rightarrow 2^-} g(x)$, $\lim_{x \rightarrow 2^+} g(x)$, $\lim_{x \rightarrow 2} g(x)$.



- (b) At which x values in $(-3, 4)$ is g not continuous? Classify each discontinuity. At which of these is g continuous from the left, or from the right?
- (c) Suppose $\lim_{x \rightarrow a} f(x) = 4$ and $\lim_{x \rightarrow a} h(x) = -2$. Evaluate $\lim_{x \rightarrow a} [3f(x) - 5h(x)]$, $\lim_{x \rightarrow a} f(x)h(x)$, $\lim_{x \rightarrow a} \frac{f(x)}{h(x)}$, and $\lim_{x \rightarrow a} \left(\sqrt{f(x)} + h(x)^2 \right)$.

3. Computing limits.

- (a) Evaluate $\lim_{x \rightarrow 3} \frac{x^2 - 2x - 3}{x^2 - 9}$.
- (b) Evaluate $\lim_{x \rightarrow 5} \frac{\sqrt{x+4} - 3}{x - 5}$.
- (c) Evaluate $\lim_{x \rightarrow -1} \frac{\frac{1}{x+3} - \frac{1}{2}}{x+1}$.
- (d) Evaluate $\lim_{x \rightarrow 4} \frac{5}{x^2 - 3x - 4}$, considering the one-sided limits.
- (e) Evaluate $\lim_{x \rightarrow 2} \frac{x^2 - 4}{|x - 2|}$.

4. Squeeze Theorem.

- (a) Evaluate $\lim_{x \rightarrow 0} x^2 \sin\left(\frac{3}{x}\right)$.
- (b) If $2x - 1 \leq f(x) \leq x^2 - 2x + 3$ for all x , find $\lim_{x \rightarrow 2} f(x)$.

5. Limits at infinity.

- (a) Evaluate $\lim_{x \rightarrow -\infty} \frac{\sqrt{4x^2 + 1}}{x + 3}$.

(b) Evaluate $\lim_{x \rightarrow \infty} (\sqrt{x^2 + 6x} - x)$.

(c) Find all vertical and horizontal asymptotes of $f(x) = \frac{2x^2 - 2}{x^2 + x - 2}$. Why is $x = 1$ not a vertical asymptote?

6. Continuity and the IVT.

(a) Let $f(x) = \begin{cases} x + 2 & x < -1 \\ x^2 & -1 \leq x \leq 1 \\ 3 - x & x > 1 \end{cases}$. Is f continuous at $x = -1$ and at $x = 1$? Where is f not differentiable?

(b) Find a and b so that $f(x) = \begin{cases} 2 & x < 1 \\ ax + b & 1 \leq x < 4 \\ 8 & x \geq 4 \end{cases}$ is continuous everywhere.

(c) Show that $x^3 + x - 1 = 0$ has a solution in $(0, 1)$.

7. The derivative as a limit.

(a) Let $f(x) = x^3$ and $P = (2, 8)$. Find the slope of the tangent line at P as a limit of slopes of secant lines through P and $Q = (x, x^3)$, then write the tangent line.

(b) Compute $f'(x)$ directly from $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ for $f(x) = \frac{x}{x+2}$.

(c) A ball's height is $h(t) = 200 - 16t^2$ feet after t seconds. Find the slope of the secant line between $t = 1$ and $t = 3$, then the instantaneous velocity at $t = 2$ as a limit.

(d) Show that $f(x) = \sqrt[3]{x}$ is continuous at $x = 0$ but not differentiable there. What does the tangent line at $x = 0$ look like?

(e) Each limit is $f'(a)$ for some f and a . Find f and a , then evaluate: $\lim_{h \rightarrow 0} \frac{(2+h)^4 - 16}{h}$ and $\lim_{x \rightarrow \pi/2} \frac{\cos x}{x - \frac{\pi}{2}}$.

8. Differentiation rules.

(a) Find y' if $y = \frac{x^2 + 1}{2x + 1}$.

(b) Find the points on the curve $y = x^3 - 3x^2 - 9x + 2$ where the tangent line is parallel to the line $y = 15x$.

(c) A particle has position $s(t) = t^3 - 9t^2 + 24t$ meters at time t seconds. Find $v(t)$ and $a(t)$, with units. When is the particle at rest?

9. Trig derivatives.

(a) Starting only from $\frac{d}{dx} \sin x = \cos x$ and $\frac{d}{dx} \cos x = -\sin x$, derive $\frac{d}{dx} \sec x$. Then find $\frac{d^2}{dx^2} \tan x$.

(b) Evaluate $\lim_{x \rightarrow 0} \frac{\sin(4x)}{3x}$ and $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x \sin x}$.

(c) Find y' if $y = \frac{\sec x}{1 + \tan x}$.

10. Chain rule.

- (a) Use the table to find $(f \circ g)'(1)$, $(g \circ f)'(1)$, and $(fg)'(2)$.

x	0	1	2	3
$f(x)$	2	3	0	1
$f'(x)$	-1	4	2	5
$g(x)$	1	2	3	0
$g'(x)$	3	-2	1	4

- (b) Find the tangent line to $y = \tan(\sqrt{x})$ at $x = \frac{\pi^2}{16}$.
 (c) Find the tangent line to $y = (x^2 - 3)^5$ at $x = 2$.
 (d) Find all points (x, y) with $0 \leq x < 2\pi$ where $y = 2\sin x + \cos(2x)$ has a horizontal tangent line.

11. Implicit differentiation.

- (a) For the ellipse $x^2 + 4y^2 = 16$, find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$, simplified using the equation of the ellipse.
 (b) Find the tangent line to $\sin(x + y) = 2x - 2y$ at (π, π) .
 (c) Find the tangent line to $x^3 + y^3 = 9xy$ at $(2, 4)$.
 (d) For $y^2 = x^3 - 3x + 2$, find $\frac{dy}{dx}$, the tangent line at $(2, 2)$, and all points with a horizontal tangent line.

12. Related rates.

- (a) A 13 ft ladder leans against a wall. The bottom slides away at 2 ft/s. How fast is the top sliding down when the bottom is 5 ft from the wall?
 (b) A tank is an inverted cone with radius 3 m at the top and height 6 m. Water is pumped in at 4 m³/min. How fast is the water level rising when the water is 4 m deep?
 (c) A 6 ft tall person walks away from a 15 ft lamppost at 3 ft/s. How fast is the tip of the person's shadow moving?
 (d) A marble moves along $x^2 + xy + y^2 = 9$. At $(3, 0)$, x is decreasing at 2 ft/s. Is y increasing or decreasing, and how fast?

13. Linear approximation.

- (a) Find the linearization of $f(x) = \sqrt[3]{x}$ at $a = 27$. Use it to approximate $\sqrt[3]{27.27}$.
 (b) Find the linearization of $f(x) = \sin x$ at $a = \frac{\pi}{6}$. Use it to approximate $\sin\left(\frac{\pi}{6} + 0.02\right)$.
 (c) The radius of a sphere is measured as 10 cm with a possible error of 0.05 cm. Estimate the maximum error in the computed volume.

14. Extreme values.

- (a) Find the critical numbers of $f(x) = x^{2/3}(x - 5)$. Which come from $f'(x) = 0$ and which from $f'(x)$ undefined?
 (b) Find the absolute max and min of $f(x) = x^4 - 8x^2 + 3$ on $[-1, 3]$.
 (c) Find the absolute max and min of $f(x) = x\sqrt{4 - x^2}$ on $[-1, 2]$.

(d) Find the absolute max and min of $f(x) = 2\cos x + \sin(2x)$ on $[0, \pi]$.

15. Mean Value Theorem.

(a) Find every c guaranteed by the MVT for $f(x) = x^2 - 4x + 1$ on $[1, 5]$.

(b) If $f(2) = -1$ and $f'(x) \leq 3$ for all x , how large can $f(6)$ be?

(c) Show that $x^3 + 5x - 2 = 0$ has exactly one real solution.

16. Shape of a graph.

(a) For $f(x) = x^3 - 6x^2 + 9x + 1$, find the intervals of increase and decrease, local extrema, concavity, and inflection points.

(b) Do the same for $f(x) = 3x^4 - 4x^3$.

(c) Find constants a and b so that $f(x) = x^3 + ax^2 + bx$ has an inflection point at $x = 1$ and a local minimum at $x = 3$. Where is its local maximum?

17. Curve sketching.

(a) Sketch $f(x) = \frac{1}{x^2+3}$, labeling intercepts, asymptotes, extrema, and inflection points.

(b) Sketch $f(x) = \frac{x^2}{x-1}$, labeling all key features.

(c) Sketch $f(x) = x - 2\sqrt{x}$, labeling all key features.

18. Optimization.

(a) A closed box has a rectangular base whose length is twice its width, and its volume is 72 ft^3 . Find the dimensions that use the least material.

(b) A closed cylindrical can holds $16\pi \text{ cm}^3$. Find the radius and height that minimize its surface area.

(c) A rectangle has its base on the x -axis and its two upper corners on the semicircle $y = \sqrt{25 - x^2}$. Find the largest possible area.

(d) A rectangular pen is split in half by a divider parallel to one side. The outer fence costs \$4 per meter and the divider \$2 per meter. With \$200, what is the largest possible area?

(e) Find the points on the parabola $y = x^2$ closest to the point $(0, 2)$.

19. Antiderivatives.

(a) Find the most general antiderivative of $g(x) = \frac{x^2-3x+2\sqrt{x}}{x}$.

(b) If $f'(x) = 3\sqrt{x} - 2$ and $f(4) = 10$, find $f(x)$.

(c) A particle has $a(t) = 6t - 4$, $v(0) = 3$, $s(0) = 1$. Find $s(t)$.

(d) Let F be the antiderivative of $f(x) = x^2 - 4$ with $F(0) = 0$. Without finding F , decide where F is increasing, where it has local extrema, and where it is concave up. Then find F .

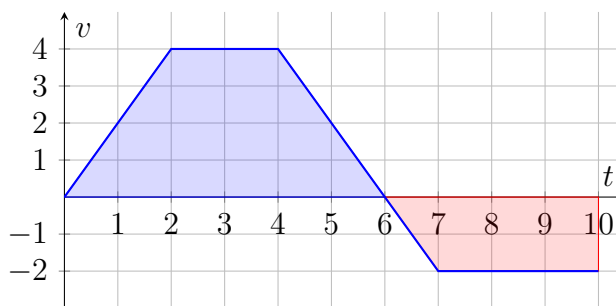
20. Riemann sums.

(a) Estimate $\int_0^\pi \sin x \, dx$ with the midpoint rule and $n = 3$.

- (b) Approximate the area under $f(x) = x^2$ on $[0, 4]$ with $n = 4$ using right endpoints, left endpoints, and midpoints. Which are over and which are under?
- (c) Write $\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{3}{n} \sqrt{1 + \frac{3i}{n}}$ as a definite integral, then evaluate it.

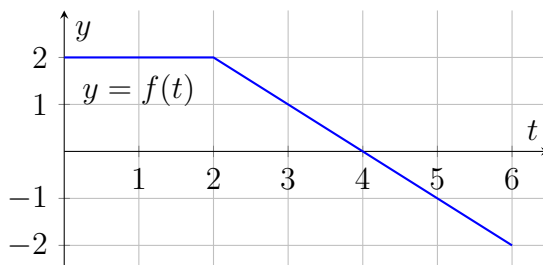
21. The integral as area.

- (a) Evaluate $\int_0^2 (3x + \sqrt{4 - x^2}) dx$ with geometry.
- (b) Evaluate $\int_{-2}^2 (x^3 + \sqrt{4 - x^2}) dx$ with geometry.
- (c) Evaluate $\int_{-3}^3 (|x| - 1) dx$ with geometry.
- (d) For the velocity graph below (v in m/s, t in s), find the displacement and total distance from $t = 0$ to $t = 10$, the position at $t = 10$ if $x(0) = 3$ m, the acceleration at $t = 5$, and when the particle is farthest from its start.



22. Fundamental Theorem, Part 1.

- (a) Find $g'(x)$ if $g(x) = \int_{x^2}^4 t \sqrt{1 + t^2} dt$.
- (b) Find $F'(x)$ if $F(x) = \int_0^{\tan x} \sqrt{1 + t^2} dt$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$, and simplify.
- (c) Find $g'(x)$ if $g(x) = \int_{2x}^{x^3} \frac{1}{1+t^4} dt$.
- (d) Let $g(x) = \int_0^x f(t) dt$, where f is graphed below. Find $g(4)$, $g'(4)$, $g''(4)$, and where on $[0, 6]$ g is largest.



23. Fundamental Theorem, Part 2.

- (a) Evaluate $\int_1^4 (3\sqrt{x} - \frac{2}{x^2}) dx$.
- (b) Evaluate $\int_0^\pi (\sin x + 2 \cos x) dx$.
- (c) What is wrong with $\int_{-1}^2 \frac{1}{x^2} dx = \left[-\frac{1}{x}\right]_{-1}^2 = -\frac{3}{2}$?

24. Substitution.

- (a) Evaluate $\int \sec^2 x \tan^3 x dx$.

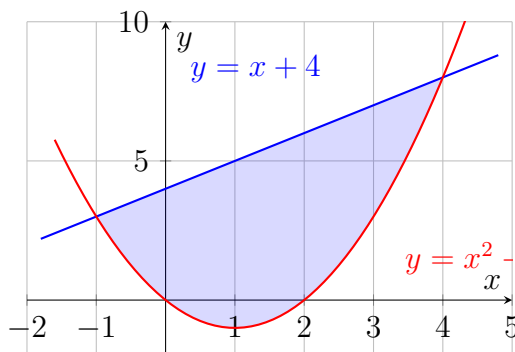
(b) Evaluate $\int_0^{\pi/2} \cos x \sqrt{1 + 3 \sin x} \, dx$.

(c) Evaluate $\int_{-2}^1 x \sqrt{x+3} \, dx$.

(d) Evaluate $\int_{-\pi/2}^{\pi/2} x^2 \sin x \, dx$.

25. Area between curves.

(a) Sketch the region between $y = x^2 - 2x$ and $y = x + 4$ and find its area.



(b) Find the area between $y = \sqrt{x}$ and $y = \frac{x}{3}$, integrating with respect to x , then with respect to y .

(c) Find the area between $y = \sin x$ and $y = \cos x$ for $0 \leq x \leq \frac{\pi}{2}$.

26. Average value.

(a) Find the average value of $f(x) = \frac{1}{x^2}$ on $[1, 3]$.

(b) Find the average value of $f(x) = x^2$ on $[0, 3]$ and a c in $[0, 3]$ with $f(c)$ equal to it.

27. Volumes.

(a) Find the volume when the region between $y = x$ and $y = x^2$ is revolved around the x -axis.

(b) Find the volume when the region under $y = \sqrt{x}$ for $0 \leq x \leq 4$ is revolved around the x -axis.

(c) Revolve the region between $y = \sqrt{x}$ and $y = x^2$ around the y -axis, and then around the line $x = -1$.

28. Work.

(a) A spring has natural length 20 cm, and 0.9 J of work stretches it from 20 cm to 26 cm. How much work stretches it from 24 cm to 28 cm? What force holds it at a length of 30 cm?

(b) A spring has natural length 10 cm and spring constant k . Compare the work W_1 from 10 cm to 15 cm with W_2 from 15 cm to 20 cm.

(c) A 10 m chain with mass 2 kg per meter hangs from the top of a building. How much work does it take to pull the whole chain up to the top? Use $g = 9.8 \text{ m/s}^2$.

(d) A force of $F(x) = \frac{x}{(x^2+1)^2}$ pounds acts on an object x feet from the origin. Find the work done moving the object from $x = 0$ to $x = 1$.